

BRANCH DOCUMENT — IRIS-3

Branch: branch_20260512_133015 | Base: main-v1.0

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Decision: GO_WITH_CONSTRAINTS | Confidence: 78%

CHANGE REQUEST

modifica il materiale mantenendo le stesse dimensioni ma facendolo tutto di marmo

CASCADE ISSUES IDENTIFIED

[CRITICAL] ISSUE-001 — Thermal Gradient Exceeds Limit

Subsystem: thermal | Req: L1-THM-001 | Type: constraint_violation

Marble has a higher thermal conductivity than typical materials used in spacecraft, which could increase the thermal gradient across the optical bench beyond the 15°C limit.

[MAJOR] ISSUE-002 — Shock Test Required

Subsystem: mechanical | Req: L1-MEC-003 | Type: missing_test

Marble is brittle and may not withstand the shock loads during launch, necessitating a new shock test to validate its structural integrity.

[MINOR] ISSUE-003 — Mass Increase

Subsystem: structure | Req: None | Type: reanalysis_required

Marble is denser than typical spacecraft materials, leading to an increase in the total mass of the payload. This could violate the 35 kg mass budget.

[MINOR] ISSUE-004 — Solar Panel Mounting Issues

Subsystem: solar | Req: None | Type: reanalysis_required

Marble may not provide adequate mounting points or structural support for the solar panels, potentially affecting their deployment and alignment.

[MINOR] ISSUE-005 — Stray Light Concerns

Subsystem: startracker | Req: None | Type: reanalysis_required

Marble's reflective properties could introduce stray light into the star tracker, affecting its performance and accuracy.

MISSING ECSS TESTS — NOW REQUIRED

No additional tests required.

SUBSYSTEM STATUS — UPDATED

Subsystem	Score	Status	Detail	Changed
optical	92%	OK	MTF 0.42 > threshold 0.40	—
thermal	61%	WARN	ΔT 14.8°C — margin < 5%	—
mechanical	88%	OK	1st mode 112 Hz > 80 Hz req	—
structure	91%	OK	Mass 33.8 kg / budget 35 kg	—
solar	89%	OK	Power margin +18%	—
startracker	67%	WARN	Stray light analysis pending	—
dataproc	71%	WARN	Compression ratio TBC	—

AFFECTED REQUIREMENTS

No requirements directly invalidated.